Problem Set 1 - Macroeconomics

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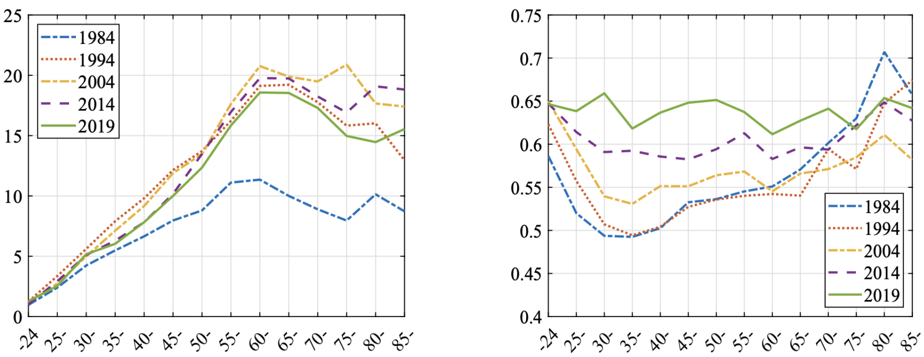
Literature review



(e) Fin. Wealth (JPY mm) (f) Fin. Wealth Gini

Fig. 10 Life-cycle Profiles: Household

Figure 1: Comparative evolution of financial wealth levels, and of the financial wage gini coefficient between 1984 and 2019, as a function of age

This paper describes the evolution of earnings, income, and wealth inequalities in Japan at the household level from the mid-1980s to 2019. The left-hand graph illustrates the trajectory of financial wealth among Japanese adults by age cohort. A substantial increase in financial wealth is evident for individuals aged 40 and above between 1984 and 2004, though this growth was partially dampened by the 2008 financial crisis. Younger demographics experienced a similar but considerably more modest upward trend. The right-hand graph depicts the evolution of the financial wealth Gini coefficient across age groups. Notably, while the financial wealth levels of 30-40 year-old Japanese remained relatively stable after 1994, the Gini coefficient for this cohort surged by approximately 0.13 points. Conversely, the age groups (50-70yo) who experienced the most important rise in financial wealth since the 1980s faced a more moderate gini coefficient increase (+0.8). Thus, while the strength of the gini coefficient was negatively correlated with age for those aged 30 and above until 1994, the heterogeneous evolution of the gini coefficient after 1994 led a marked decrease of the relation between age and inequality levels. This points to an ambiguous effect of the asset price bubble collapse of the 1990s which seems to have been associated with both an increase in financial wealth across cohorts and a leveling up of wealth inequalities for all age groups, save for senior Japanese.

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Exercise I

2.1 1)

The problem : k_0 given and $k_t, c_t \geq 0$

$$\max_{\{k_t,c_t\}_0^\infty} \sum_{t=0}^{\infty} \beta^t log(c_t) \quad \text{s.t } c_t + k_{t+1} \leq \theta k_t^\alpha$$

At the optimum, $c_t + k_{t+1} = \theta k_t^\alpha$ Hence the problem is :

$$\max_{\{k_t\}_0^\infty} \sum_{t=0}^{\infty} \beta^t log(\theta k_t^\alpha - k_{t+1})$$

In recursive form, we get :

$$V(k_0) = \max_{0 \leq k' \leq \theta k'^{\alpha}} \{log(\theta k^{\alpha} - k') + \beta V(k')\}$$

We denote the feasibility set $\Gamma(k') = [0; \theta k^{\alpha}]$ for k'. k is a state variable, k' is a control varaible.

2.2 2)

The FOC yields, with respect to k' :

$$\frac{-1}{\theta k^{\alpha} - k'} + \beta V'(k') = 0 \Longleftrightarrow \beta(\theta k^{\alpha} - k')V'(k') = 1$$
$$\Longleftrightarrow \beta V'(k') = \frac{1}{\theta k^{\alpha} - k'}$$

The envelope theorem yields the condition :

$$V'(k) = \frac{\theta \alpha k^{\alpha-1}}{\theta k^{\alpha} - k'}$$

Hence, $V'(k') = \frac{\theta \alpha k'^{\alpha-1}}{\theta k'^{\alpha} - k''}$ Thus we derive the Euler equation :

$$\beta \frac{\theta \alpha k'^{(\alpha-1)}}{\theta k'^{\alpha} - k''} = \frac{1}{\theta k^{\alpha} - k'} \Longleftrightarrow \alpha\beta\theta k'^{\alpha-1} = \frac{\theta k'^{\alpha} - k''}{\theta k^{\alpha} - k'}$$

2.3 3)

2.3.1 a)

We guess that $\forall x \in \Gamma, V_0(x) = 0$. We have :

$$V_1(k) = TV_0(k) = \max_{k' \in \Gamma(k)} log(\theta k^{\alpha} - k') + \beta \underbrace{V_0(k')}_{=0}$$
$$= log(\theta k^{\alpha})$$

2.3.2 b)

We get :

$$V_2(k) = TV_1(k) = \max_{k' \in \Gamma(k)} log(\theta k^{\alpha} - k') + \beta \underbrace{log(\theta k'^{\alpha})}_{=V_1(k')}$$

Let $f(k') = log(\theta k^{\alpha} - l') + \beta log(\theta k'^{\alpha})$. We derive its FOC with respect to k' :

$$f'(k) = \frac{-1}{\theta k^{\alpha} - k'} + \frac{\beta\alpha}{k'} = 0$$
$$\Longleftrightarrow k' = \alpha\beta(\theta k^{\alpha} - k')$$
$$\Longleftrightarrow (1 + \alpha\beta)k' = \alpha\beta\theta k^{\alpha}$$
$$\Longleftrightarrow k' = \underbrace{\frac{\alpha\beta}{1 + \alpha\beta}}_{\leq 1} \theta k^{\alpha} \leq \theta k^{\alpha}$$

Thus, we have found the optimal k', so :

$$V_2(k) = log(\theta k^{\alpha} - \frac{\alpha\beta}{1 + \alpha\beta}\theta k^{\alpha}) + \beta log([\frac{\theta\beta\alpha k^{\alpha}}{1 + \beta\alpha}]^{\alpha}) + \beta log(\theta)$$
$$= log(\frac{1}{1 + \alpha\beta}) + \alpha\beta log(\frac{\theta\beta\alpha k^{\alpha}}{1 + \beta\alpha}) + \beta log(\theta)$$
$$= \alpha(1 + \alpha\beta)log(k) + [(1 + \beta + \alpha\beta)log(\theta) + \alpha\beta log(\alpha\beta) - (1 + \alpha\beta)log(1 + \alpha\beta)]$$

2.4 4)

Let's now assume $V(k) = a_1 + a_2 log(k)$, we have :

$$a_1 + a_2 log(k) = \max_{k' \in \Gamma(k)} log(\theta k^\alpha - k') + \beta(a_1 + a_2 log(k'))$$

Let $g(k') = log(\theta k^\alpha - k') + a_1\beta + a_2\beta log(k')$ We have :

$$\begin{aligned} g'(k') = 0 &\iff \frac{1}{\theta k^\alpha - k'} = \frac{a_2\beta}{k'} \\ &\iff k' = \theta\beta a_2 k^\alpha - a_2\beta k' \\ &\iff k' = \frac{\theta\beta a_2}{1 + a_2\beta} k^\alpha \end{aligned}$$

So we get :

$$\begin{aligned} a_1 + a_2 log(k) &= log(\theta k^\alpha(1 - \frac{a_2\beta}{1+a_2\beta})) + a_1\beta + a_2\beta log(\theta k^\alpha \frac{\beta a_2}{1+a_2\beta}) \\ &= log(\theta\frac{1}{1+a_2\beta}) + \alpha log(k) + a_1\beta + a_2\beta log(\theta\frac{\theta a_2\beta}{1+a_2\beta}) + \alpha a_2\beta log(k) \end{aligned}$$

Hence we get :

$$a_2 = \alpha + \alpha a_2\beta \iff a_2 = \frac{\alpha}{1-\alpha\beta}$$

Then, we find a_1 :

$$\begin{aligned} a_1 &= a_1\beta + log(\frac{\theta}{1+\frac{\alpha\beta}{1-\alpha\beta}}) + \frac{\alpha\beta}{1-\alpha\beta} log(\theta\frac{\frac{\alpha\beta}{1-\alpha\beta}}{1+\frac{\alpha\beta}{1-\alpha\beta}}) \\ \iff a_1(1-\beta) &= log(\theta(1-\alpha\beta)) + \frac{\alpha\beta}{1-\alpha\beta} log(\theta\alpha\beta) \\ \iff a_1 &= \frac{1}{1-\beta}[log(\theta(1-\alpha\beta)) + \frac{\alpha\beta}{1-\alpha\beta} log(\theta\alpha\beta)] \end{aligned}$$

2.5 5)

Let's find the policy functions : Based on (4) we get :

$$\begin{aligned} k' &= \frac{\beta a_2 \theta k^\alpha}{1+\beta a_2} \\ &= \frac{\beta(\frac{\alpha}{1-\beta\alpha})\theta k^\alpha}{1+\beta(\frac{\alpha}{1-\alpha\beta})} \\ &= \beta\alpha\theta k^\alpha \end{aligned}$$

So we have :

$$k' = \beta\alpha\theta k^\alpha \equiv g'_k(k)$$

Similarly, we get the policy function for c :

$$c = \theta k^\alpha - k' = \theta k^\alpha - \beta\alpha\theta k^\alpha = (1-\beta\alpha)\theta k^\alpha \equiv g_c(k)$$

2.6 6)

Let us take $k_0 = 1, \alpha = \frac{1}{2}, \beta = 0.9, \theta = 2$, the we get :

$$\begin{aligned} k_1 &= 0.9 \\ k_2 &= \frac{9}{10} \times \frac{9}{10}^{1/2} \approx 0.854 \\ k_3 &= \frac{9}{10}^{5/2} \approx 0.768 \end{aligned}$$

And for the consumption path we get :

$$\begin{aligned}
c_0 &= \theta k_0^\alpha - k_1 = 2 - 0.9 = 1.1 \\
c_1 &= \theta k_1^\alpha - k_2 = 2 \times 0.9^{1/2} - 0.854 \approx 1.0436 \\
c_2 &= \theta k_2^\alpha - k_3 \approx 0.985
\end{aligned}$$

3 Exercise 2

3.1 1)

- We assume that housing goods can be purchased at each period t for the corresponding price P_t. Household take the latter as granted. Purchased housing goods then depreciate at rate δ and can be sold at the price P_t to acquire additional resources.
- it is assumed that there is no market friction p : 1- the agent has perfect information about the price of housing goods in every period) and full ownership of the good is transferred instantaneously (can sell it back at price P_t at the next period, accounting for depreciation).

3.2 2)

To avoid Ponzi schemes, we need to impose the following transversality condition on the financial asset b_t :

$$\lim_{t \longrightarrow +\infty} \beta^t b_t \geq 0$$

Similarly, for the amount of housing good H_t, we need :

$$\lim_{t \longrightarrow +\infty} \beta^t P_t H_t \geq 0$$

3.3 3)

Let us assume that P_t, y_t and r_t are parameters and are determined exogenously at each period. Households choose how much of the housing good H_t they wish to invest in, how much they wish to consume now c_t and how much they wish to save for the next period through the financial asset b_t. The choice of the control variables c_t, H_t, b_t at each period depends on parameters and on the available stock of housing goods H_{t-1} and financial assets b_{t-1}.
Now that we have identified the state and control variable, we can write the value function through the Bellman equation :

$$V(b_{-1}, H_{-1}) = \max_{H,b} u(y + (1 - r_{-1})b_{-1} + P(1 - \delta)H_{-1} - b - PH, H) + \beta V(b, H)$$

3.4 4)

We begin by deriving the FOC in the control variables H, b :

$$\begin{aligned}
\frac{\delta V(b_{-1}, H_{-1})}{\delta b} &= 0 \Longleftrightarrow u_1(c, H) = \beta V_1(b, H) \\
\frac{\delta V(b_{-1}, H_{-1})}{\delta H} &= 0 \Longleftrightarrow P u_1(c, H) - u_2(c, H) = \beta V_2(b, H)
\end{aligned} \tag{1}$$

Then the envelope theorem yields the following conditions in H_{-1}, b_{-1} (state variables):

$$\begin{aligned}
V_1(b_{-1}, H_{-1}) &= (1 - r_{-1}) u_1(c, H) \\
V_2(b_{-1}, H_{-1}) &= P(1 - \delta) u_1(c, H)
\end{aligned} \tag{2}$$

Let's substitute (2) in (1) after evaluating the former at (b, H) :

$$\begin{aligned} u_1(c,H) &= \beta(1-r)u_1(c',H') \\ Pu_1(c,H) + u_2(c,H) &= \beta P'(1-\delta)u_1(c',H') \end{aligned} \tag{3}$$

Thus we get the Euler equations :

$$\begin{aligned} u_1(c,H) &= \beta(1-r)u_1(c',H') \\ u_2(c,H) &= \beta P'(1-\delta)u_1(c',H') - Pu_1(c,H) \end{aligned} \tag{4}$$

3.5 Bonus question

Let's use (4) to get $u_1(c', H')$ as a function of $u_1(c, H)$:

$$u_1(c',H') = \frac{u_1(c,H)}{\beta(1-r)}$$

Now, the second equation in (4) gives us :

$$\begin{aligned} u_2(c,H) &= \beta P' \frac{1-\delta}{\beta(1-r)} u_1(c,H) - Pu_1(c,H) \\ &= (P' \frac{1-\delta}{1-r} - P)u_1(c,H) \end{aligned}$$

Hence, with the notation in t, t+1 :

$$u_2(c_t,H_t) = (P_{t+1}\frac{1-\delta}{1-r_t} - P_t)u_1(c_t,H_t) \tag{5}$$

Let us compute $u_2(c_t, H_t)$ and $u_1(c_t, H_t)$:

$$\begin{cases} u_1(c_t,H_t) = c_t^{\rho-1}(c_t^\rho + H_t^\rho)^{\frac{1}{\rho}-1} \\ u_2(c_t,H_t) = H_t^{\rho-1}(c_t^\rho + H_t^\rho)^{\frac{1}{\rho}-1} \end{cases}$$

Let's denote $U \equiv H_t^{\rho-1}(c_t^\rho + H_t^\rho)^{\frac{1}{\rho}-1}$. We substitute $u_2(c_t, H_t)$ and $u_1(c_t, H_t)$ in (5) and express $\frac{c_t}{H_t}$ as a function of other variables :

$$\begin{aligned} H^{\rho-1}U &= (P_{t+1}(\frac{1-\delta}{1-r_t}) - P_t)c_t^{\rho-1}U \\ \iff (\frac{c_t}{H_t})^{\rho-1} &= (P_{t+1}(\frac{1-\delta}{1-r_t}) - P_t)^{-1} \\ \iff (\frac{c_t}{H_t}) &= (P_{t+1}(\frac{1-\delta}{1-r_t}) - P_t)^{\frac{1}{1-\rho}} \end{aligned}$$

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Exercise 3

Please find the associated Pluto Notebook at the following link : Notebook for Exercise 3

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Reference

Kitao, S., Yamada, T. Earnings, income, and wealth inequality in Japan: a long-term perspective, 1984–2019. JER 76, 231–283 (2025). https://doi.org/10.1007/s42973-024-00187-0